IMPACT : COLLISION OF ELASTIC BODIES :

MSIT

* ELASTIC BODIES : The bodies which regain their original shape to less or more extent are called elastic bodies.

* INELASTIC / PLASTIC BODIES : The body which does not regain at all.

* COLLISION : means contact between two bodies for a short duration.

* IMPACT : The phenomena of collision between two bodies which occur in a very short interval of time and during which the bodies exert relatively large forces on each other is called an IMPACT.

* Restitution : Just after impact the bodies attempt to regain their original shape, due to their elasticity. This process, of regaining its original shape is called restitution.

* SOME IMPORTANT TERMS RELATED TO COLLISION :

i> Time of Compression / Deformation : The time taken by two bodies in compression / Deformation after the instant of collision, is known as time of compression.

ii> Time of Restitution : The time taken by two bodies to regain the original shape after deformation is known as time of restitution.

iii> Time of Collision : The sum of time of compression and time of restitution is known as time of collision or period of collision or period of impact.

* TYPES OF IMPACT :

→ LINE OF IMPACT : The line joining the centres of the colliding bodies and passing through the point of contact is called the line of impact for central impact. It represents the common normal to the surfaces of the bodies in contact and passes through the point at which collision takes place.

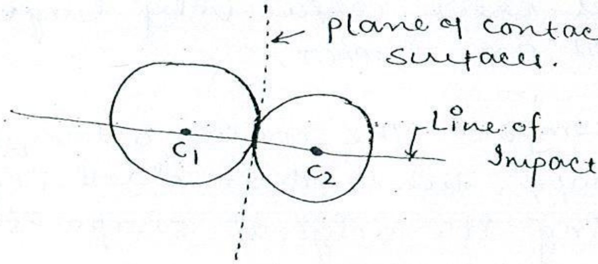
When two bodies collide with one another, the result impact may be

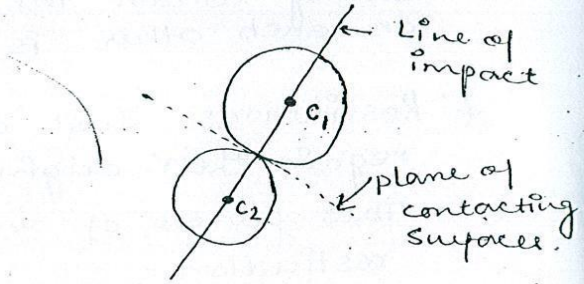
i) Central and eccentric impact

ii) Direct and oblique impact

- Central Impact: The impact is called central when the mass centres of colliding bodies are located on the line of impact.

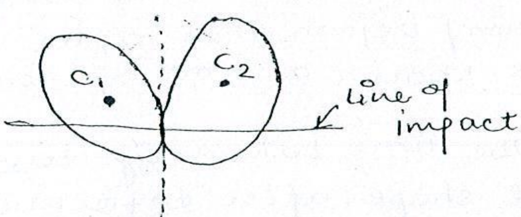
  The fig. given below represents central impact; the mass centres C_1 and C_2 of colliding bodies lies on the line of impact.

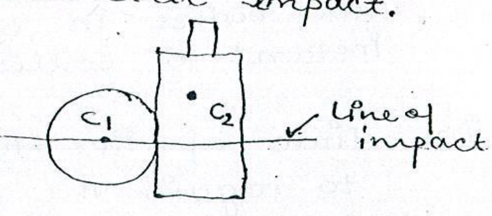




- Eccentric Impact: when the mass centres of the colliding bodies are not located on the line of impact, the impact is referred to as eccentric impact.

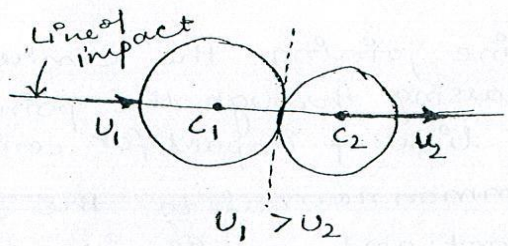
Eg:

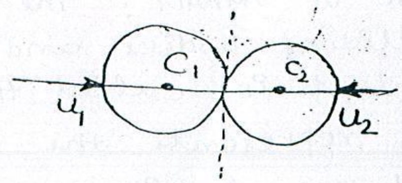




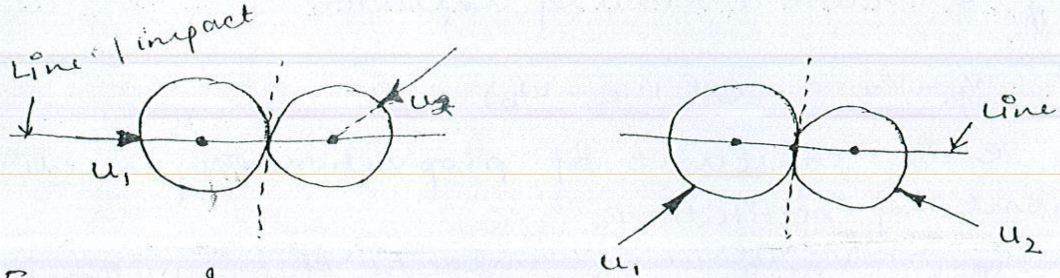
- Direct Impact: The impact is said to be direct if before impact the bodies are moving along the line of impact, i.e. motion of colliding bodies is directed along the line of impact.

  For example:





- Indirect / Oblique Impact: If the motion of one or both of the colliding bodies, before impact, is not directed along the line of impact.



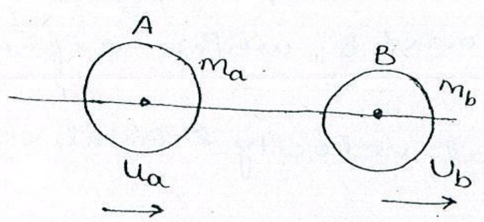
* ELASTIC IMPACT: The impact is elastic if the body rebounds after impact.

* INELASTIC IMPACT: The impact is inelastic if the bodies taking part in collision does not rebound at all

* NEWTON'S LAW OF COLLISION: COEFFICIENT OF RESTITUTION

According to this law when two moving bodies collide with each other, their velocity of seperation bear constant ratio to their velocity of approach.

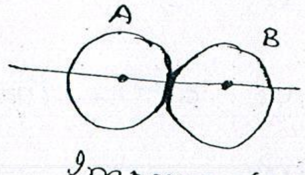
Consider two bodies A and B of mass m_a and m_b resp. Let these bodies be moving with velocity u_a and u_b before impact. The impact will take place only if $u_a > u_b$



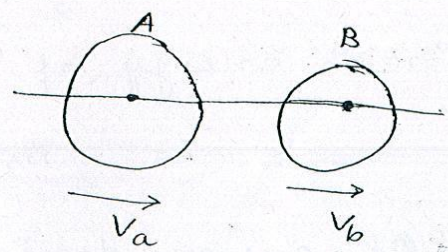
Then velocity of approach $= u_a - u_b$

Before Impact

After a short period of contact the bodies will seperate and start moving with velocities V_a and V_b resp.



Impact (maximum deformatn)



The seperation will occur only if $V_b > V_a$

∴ velocity of seperation $= v_b - v_a$

According to Newtons Law of Collision

$$v_b - v_a = e\,(u_a - u_b)$$

where e is constant of proportionality and is called coefficient of restitution.

$$e = \frac{v_b - v_a}{u_a - u_b} = \frac{\text{velocity of seperation}}{\text{velocity of approach.}}$$

The coefficient of restitution is a parameter which indicates the energy loss during impact.

- If $e = 0$, bodies are inelastic i.e. there is no restitution.
- If $e = 1$, the bodies are perfectly elastic i.e. they fully regain their original shape.

The value of coefficient of restitution depends on
- material of body
- shape of the body and
- size of the body.

→ DIRECT IMPACT:

* LOSS OF KINETIC ENERGY DURING DIRECT CENTRAL Impact:

Consider two bodies A and B which experience direct impact. Let

m_1, u_1, v_1 = mass, initial velocity & final velocity of body A

m_2, u_2, v_2 = mass, initial velocity & final velocity of body B.

The kinetic energy of two masses before impact

$$= \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

Kinetic energy of two masses after impact

$$= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Loss of energy during impact.

$$\Delta E = \frac{1}{2}\left((m_1 u_1^2 + m_2 u_2^2) - (m_1 v_1^2 + m_2 v_2^2)\right)$$

$$\Rightarrow \Delta E = \frac{1}{2(m_1+m_2)}\left[m_1 m_2 (u_1-u_2)^2 - m_1 m_2 (v_1-v_2)^2\right]$$

or $$\Delta E = \frac{m_1 m_2}{2(m_1+m_2)}(u_1-u_2)^2(1-e^2)$$

e is coefficient of restitution

- when $e=0$ (inelasic impact)

$$\Delta E = \frac{m_1 m_2}{2(m_1+m_2)}(u_1-u_2)^2$$

- when $e=1$ (perfectly elastic impact)

$$\Delta E = \frac{m_1 m_2}{2(m_1+m_2)}(u_1-u_2)^2(1-1) = 0$$

which shows

Kinetic energy before impact = Kinetic energy after impact.

∴ In case of perfectly elastic impact both momentum and kinetic energy are conserved.

IMPORTANT CASES OF IMPACT :-

- ~~Perfect~~ Plastic impact ($e=0$)

$$e = 0 = \frac{v_b - v_a}{u_a - u_b} \Rightarrow v_b = v_a = v$$

i.e. final velocities of bodies become equal after the impact.

- ELASTIC IMPACT

a) Impact of two equal masses:

$$m_a = m_b = m$$

According to Law of Conservation of momentum

$$m_a u_a + m_b u_b = m_a v_a + m_b v_b$$

$$u_a + u_b = v_a + v_b \quad -(I)$$

Also $e = \dfrac{v_b - v_a}{u_a - u_b} = 1$ gives $v_b - v_a = u_a - u_b$ —(II)

Solving (I) & (II)

$$v_a = u_b \quad \text{and} \quad v_b = u_a$$

i.e. The masses exchange velocities after impact.

b) Impact of two bodies in which one is immovable and very large mass as compared to other.

$m_b = \infty$

$V_b = 0$

According to law of conservation of momentum

$$m_a V_a + m_b V_b = m_a V_a' + m_b V_b'$$

Dividing by m_b

$$\frac{m_a}{m_b} V_a + \frac{m_b}{m_b} V_b = \frac{m_a}{m_b} V_a' + V_b'$$

when m_b is very large and V_b is zero

$$V_b' = 0$$

c) A Body strikes another body of Equal mass at rest

$m_a = m_b = m$, $V_b = 0$

Conservation of momentum gives.

$$m_a V_a + m_a V_b = m_a V_a' + m_a V_b'$$

or $\quad V_a + 0 = V_a' + V_b'$

$$e = 1 = \frac{V_b' - V_a'}{V_a - V_b} \Rightarrow V_a - V_b = V_b' - V_a'$$

Solving the above eq^{n}s :

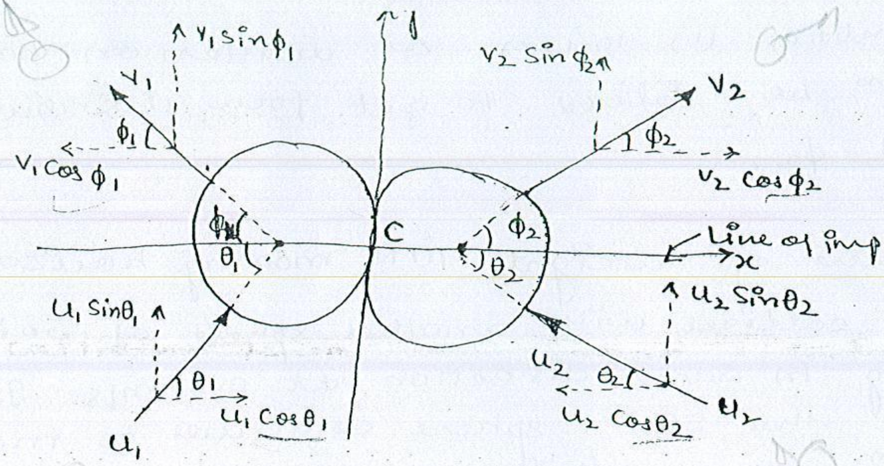
$$V_a' = 0 \quad \text{and} \quad V_b' = V_a$$

The striking mass stops after imparting its entire velocity to the mass at rest.

★ OBLIQUE CENTRAL IMPACT:

Consider two spherical bodies having masses of m_1 & m_2 moving resp. with velocities u_1 and u_2 as shown in fig.

The velocities are not directed along the line of impact and as such the colliding system represents the oblique central impact.



We choose x-axis along the line of impact and y axis along the tangent line at the point of contact. Then Let

θ_1 and θ_2 are angles which initial velocities u_1 and u_2 make with the x-axis and

ϕ_1 and ϕ_2 are the angles which the final velocities v_1 & v_2 make with the x-axis.

Motion along normal to Line of impact (y-axis):

Using the principle of conservation of momentum in y-direction for each body individually

i.e. $m_1 u_1 \sin\theta_1 = m_1 v_1 \sin\phi_1$ or $u_1 \sin\theta_1 = v_1 \sin\phi_1$ —(i)

and $m_2 u_2 \sin\theta_2 = m_2 v_2 \sin\phi_2$ or $u_2 \sin\theta_2 = v_2 \sin\phi_2$ —(ii)

From here it is evident that component of velocity in y-directⁿ for each body remains unchanged.

Motion along Line of Impact (x-axis)

Using principle of conservation for the system (i.e. both bodies taken together)

i.e. $m_1 u_1 \cos\theta_1 + m_2 u_2 \cos\theta_2 = m_1 v_1 \cos\phi_1 + m_2 v_2 \cos\phi_2$ —(iii)

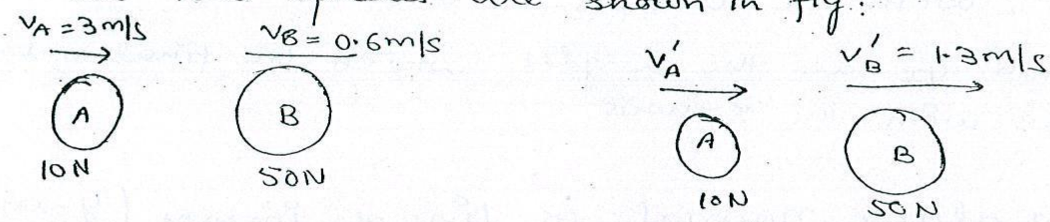
In case of oblique impact the coefficient of restitution is given by

$$e = \frac{v_2 \cos\phi_2 - v_1 \cos\phi_1}{u_1 \cos\theta_1 - u_2 \cos\theta_2} \quad \text{—(iv)}$$

The identities (i) to (iv) as written on back of the page can be solved to get four unknowns V_1, V_2, ϕ_1 and ϕ_2.

1) A sphere of weight 10 N moving horizontally at 3 m/s collides with another sphere of weight 50 N moving in same direction at 0.6 m/s. If after impact the 50 N sphere continues to move in same direct^n with a velocity of 1.3 m/s, what is the velocity of I^{st} sphere after impact?
Also determine the coefficient of restitution and the loss in kinetic energy due to impact.

Sol^n:→
The two spheres are shown in fig.:



a) Before impact   b) after impact

Applying Law of conservation of momentum we have

$$m_A V_A + m_B V_B = m_A V_A' + m_B V_B'$$

$$\frac{10}{9.81}(3) + \frac{50}{9.81}(0.6) = \frac{10}{9.81} \times V_A' + \frac{50}{9.81} \times 1.3$$

$$\Rightarrow V_A' = -0.50 = \underleftarrow{0.5 \text{ m/s}}$$ Ans.

From Newton's Law of Collision we have:

$$V_B' - V_A' = e\,(V_A - V_B)$$

$$\Rightarrow e = \frac{V_B' - V_A'}{V_A - V_B} = \frac{1.3 - (-0.5)}{3 - (0.6)}$$

$$\therefore e = 0.75$$ Ans.

K.E. Before impact $= \frac{1}{2}\left(\frac{10}{9.81}\right)(3)^2 + \frac{1}{2}\left(\frac{50}{9.81}\right)(0.6)^2$

$= 5.503$ Nm 5.5

K.E. after impact $= \frac{1}{2}\left(\frac{10}{9.81}\right)(-0.5)^2 + \frac{1}{2}\left(\frac{50}{9.81}\right)(1.3)^2$

$= 4.435$ Nm.

Hence Loss in kinetic energy due to impact

$= 5.503 - 4.435$

$= 1.068$ Nm. Ans.

2) The masses of 2 balls are in the ratio 2: 1 and before impact, their velocities in the opposite direction conform to the ratio 1:2. If the coefficient of restitution for the direct impact is 5/6 determine the velocity as fraction of original velocity with which each ball would move after impact.

Soln. Let suffix 1 & 2 denote the 1st & 2nd ball.

From Law of conservation of momentum

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Substituting the given data $m_2 = m_1/2$ and $u_2 = -2u_1$

$$m_1u_1 + \frac{m_1}{2}(-2u_1) = m_1v_1 + \frac{m_1}{2}v_2$$

$$\Rightarrow 2m_1u_1 - 2m_1u_1 = 2m_1v_1 + m_1v_2$$

$$\Rightarrow v_2 = -2v_1 \quad \text{—(i)}$$

From coefficient of restitution,

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow \frac{5}{6} = \frac{v_2 - v_1}{u_1 - (-2u_1)}$$

$$\Rightarrow v_2 - v_1 = \frac{5}{2}u_1 \quad \text{—(ii)}$$

From (i) & (ii) we have

$$-2v_1 - v_1 = \frac{5}{2}u_1 \Rightarrow v_1 = -\frac{5}{6}u_1$$

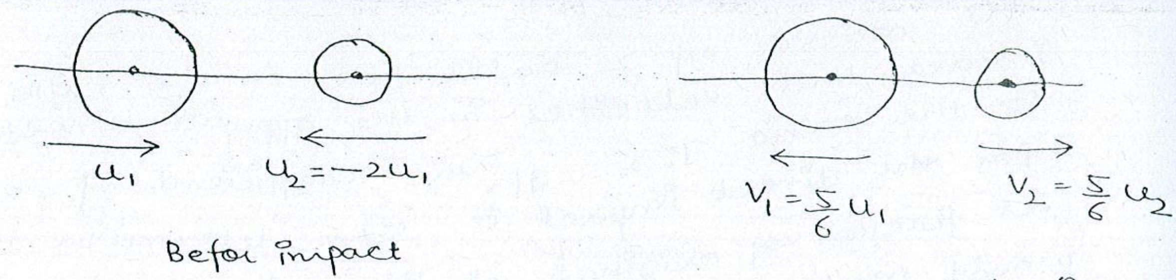
-ve sign indicates that the directn of v_1 is opposite to that of u_1. This implies that the first ball will move in opposite directn with $\frac{5}{6}$ th of its original velocity

Substituting the values of v_1 in expression (i) we have:

$$v_2 = -2v_1 = -2\left(-\frac{5}{6}u_1\right) = \left(-\frac{5}{6}\right)(-2u_1) = -\frac{5}{6}u_2.$$

$\Rightarrow$ the 2nd ball will move in opposite directn with $\frac{5}{6}$th of its original velocity.

i.e.



i.e. The directn of both the balls gets reversed after impact.

3) A Body of 10 kg mass moving towards right with a speed of 8 m/s strikes with another body of mass 20 kg moving towards left with 25 m/s. Determine if $e = 0.65$

a) Final velocity of the two bodies.

b) loss in kinetic energy due to impact and

c) Impulse acting on either body during impact.

Soln:- Let suffix 1 & 2 refer to bodies of 10 kg mass & 20 kg mass respectively.

a) From the law of conservatn of momentum

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$10\times 8 + 20(-25) = 10v_1 + 20v_2$$

$$80 - 500 = 10v_1 + 20v_2$$

$$v_1 + 2v_2 = -42 \quad \text{-(i)}$$

From the relatn of coefficient of restitution

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow 0.65 = \frac{v_2 - v_1}{8(-25)} \Rightarrow v_2 - v_1 = 21.45 \quad \text{-(ii)}$$

From eq (i) & (ii)

$$(V_1 + 2V_2) + (V_2 - V_1) = -42 + 21.45$$

$$3V_2 = -20.55$$

$$\Rightarrow V_2 = \underline{-6.85 \text{ m/s } (\leftarrow)} \quad \text{Ans.}$$

and $V_1 = V_2 - 21.45 = -6.85 - 21.45 = \underline{-28.3 \text{ m/s } (\leftarrow)}$ Ans.

i.e. Both bodies will move towards left after the impact.

b) K.E. before impact $= \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} \times 10 \times 8^2 + \frac{1}{2} \times 20 \times (-2$

$= 6570$ Nm.

K.E. after impact $= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} \times 10(-28.3)^2 + \frac{1}{2} \times 20(-6.8$

$= 4473.67$ Nm.

$\therefore$ Loss in K.E. due to impact $= 6570 - 4473.67$

$= \underline{2096.33 \text{ Nm.}}$ Ans.

c) Applying the principle of impulse-momentum we have considering body of 10kg mass

Impulse $= m_1 v_1 - m_1 u_1 = 10(-28.3) - 10 \times 8$

$= -363$ Ns.

considering body of 20 kg mass

Impulse $= m_2 v_2 - m_2 u_2 = 20(-6.85) - 20(-25)$

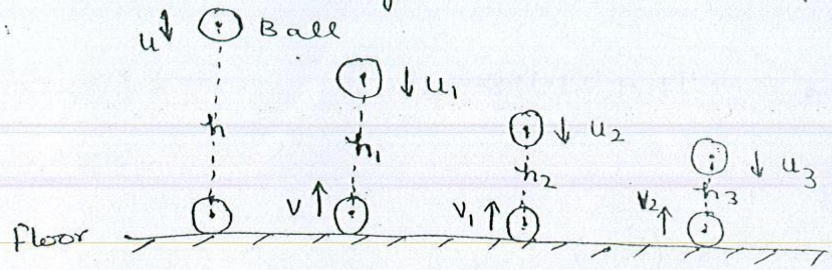
$= 363$ Ns.

$\Rightarrow$ Impulse experienced by each body is equal in magnitude but opposite in direction.

4) From what height must A heavy elastic ball be dropped on a floor so that after rebounding thrice, it will reach a height of 10m? Take coefficient of restitution $e = (0.5)^{1/3}$

Solⁿ:- Let u & v be the velocity of ball at Iˢᵗ impact (strike & bounce) with the floor.

then $u = \sqrt{2gh}$ $(\downarrow)$ & $v = \sqrt{2gh_1}$ $(\uparrow)$



The velocity of floor is zero before & after every impact i.e. $u_f = v_f = 0$

$\therefore$ from the coefficient of restitution,

$$e = \frac{v_f - v}{u - u_f} = \frac{-v}{u}$$

Since u and v are in opposite direction

$$e = \frac{v}{u} = \frac{\sqrt{2gh_1}}{\sqrt{2gh}} = \left(\frac{h_1}{h}\right)^{1/2}$$

or $h_1 = e^2 h$

Like wise: $h_2 = e^2 h_1 = e^4 h$

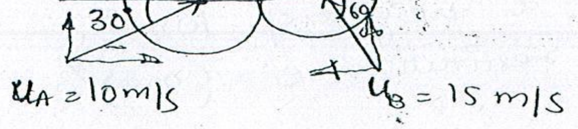
and $h_3 = e^2 h_2 = e^6 h$

Substituting the given data: $h_3 = 10\,m$ & $e = (0.5)^{1/3}$

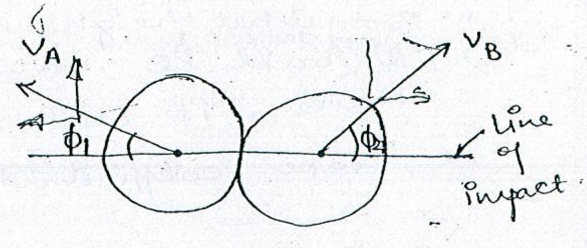
$$h = \frac{h_3}{e^6} = \frac{10}{(0.5)^2} = 40\,m.$$

Thus the ball must be dropped from a height of 40 m. Ans

5) Two identical spheres A and B strike each other as shown in fig. If $e = 0.9$ find velocity of each ball after impact.



Before impact



Sol^n : Considering motion along line of impact

$$m_A = m_B = m$$

$$m_A u_A + m_B u_B = m_A V_A + m_B V_B$$

$$m(10 \cos 30^\circ) + m(-15 (\cos 60^\circ) = m(-V_A \cos \phi_1) + m(V_B \cos \phi_2)$$

$$e = \frac{V_B - V_A}{u_A - u_B} \quad -(i)$$

$$0.9 = \frac{V_B \cos \phi_2 - (-V_A \cos \phi_1)}{(10 \cos 30^\circ) - (-15 \cdot \cos 60^\circ)} \quad -(ii)$$

considering motion perpendicular to line of impact

$$V_A \sin \phi_1 = 10 \sin 30^\circ \quad -(iii)$$

$$V_B \sin \phi_2 = 15 \sin 60^\circ \quad -(iv)$$

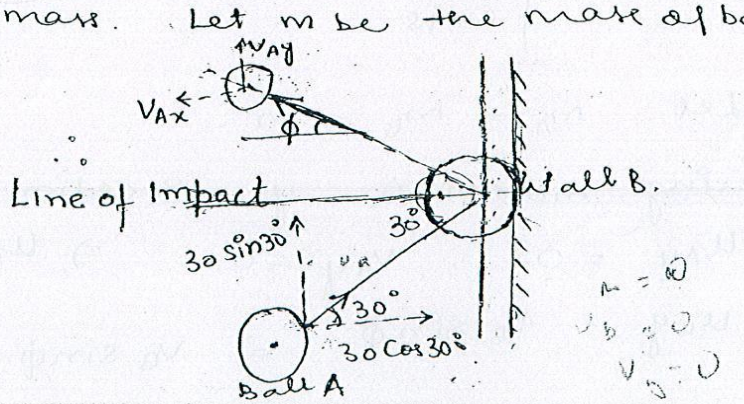
solving the above four equations

$V_A = 8.35$ m/s $\quad \phi_1 = 36.16^\circ$

$V_B = 15.2$ m/s $\quad \phi_2 = 58.84^\circ$ Ans.

6) A ball is thrown against a wall with a velocity of 30 m/s making an angle of 30° with the horizontal as shown in fig. Assuming smooth wall find the magnitude & direction of velocity of ball as it rebounds from the wall. $e = 0.50$.

Sol^n:→ Here the wall is fixed & can be considered as infinite mass. Let m be the mass of ball.



Conservatⁿ of linear momentum cannot be applied becoz of infinite mass of wall.

Consider the motion of the ball parallel to the wall. Since the ball is smooth. the component of velocity parallel to wall remains the same

$\therefore \quad u_{Ay} = V_{Ay} = 30 \sin 30° = 15 \, m/s.$

Considering motion of Ball perpendicular to the wall

$$V_{Bx} - V_{Ax} = e\,(u_{Ax} - u_{Bx})$$

$$\Rightarrow \quad 0 - V_{Ax} = 0.5\,(30 \cos 30° - 0)$$

$$\Rightarrow \quad V_{Ax} = -13 \, m/s$$

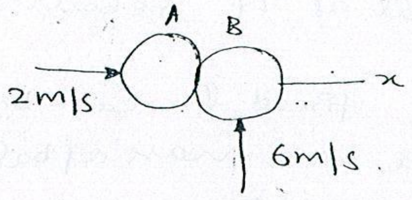
$$= 13 \, m/s \text{ towards left}$$

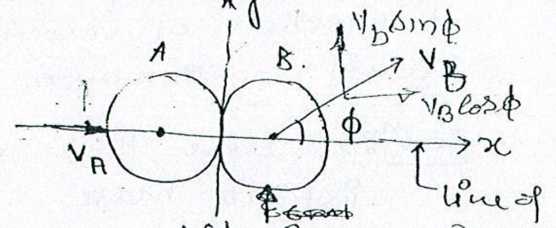
$$\therefore \quad V_A = \sqrt{(V_{Ax})^2 + (V_{Ay})^2} = \sqrt{(15)^2 + (13)^2}$$

$$= 19.8 \, m/s \quad \underline{Ans.}$$

The angle ϕ made by the ball with the x-axis given by $\tan\phi = \dfrac{15}{13}$ or $\phi = 49.08°$ Ans.

7. Two smooth identical balls strike each other as shown in fig. If $e = 0.8$ determine velocities after impact.





After impact

$\underline{Sol^n}$:→

Let $\quad m_A = m_B = m$

Considering motion in y-direction :

$u_{Ay} = 0 \qquad V_{Ay} = 0 \qquad \Rightarrow u_{Ay} = V_{Ay} = 0$

and $\quad u_{By} = V_B \sin\phi \qquad \Rightarrow \quad V_B \sin\phi = 6 \quad —(i)$

Considering motion along line of impact i.e. x- direct^n;

$$m u_A + m u_B \cos\phi = m V_A + m V_B \cos\phi$$

$$\Rightarrow u_A + u_B \cos\phi = V_A + V_B \cos\phi$$

$$\Rightarrow 2 + 0 = V_A + V_B \cos\phi$$

$$\Rightarrow V_A + V_B \cos\phi = 2 \quad \text{—(ii)}$$

From the relation of coefficient of restitution we have

$$e = \frac{V_B \cos\phi - V_A}{u_A - u_B \cos\phi} \Rightarrow \frac{V_B \cos\phi - V_A}{2 - 0} = 0.8$$

$$\Rightarrow V_B \cos\phi - V_A = 1.6 \quad \text{—(iii)}$$

Solving eq. (ii) & (iii) we get

$$V_B \cos\phi = 1.8 \quad \text{—(iv)}$$

Dividing eq. (i) by (iv) we have

$$\frac{V_B \sin\phi}{V_B \cos\phi} = \frac{6}{1.8} \Rightarrow \tan\phi = 3.33$$

$$\Rightarrow \phi = \underline{73.3^\circ} \text{ Ans}$$

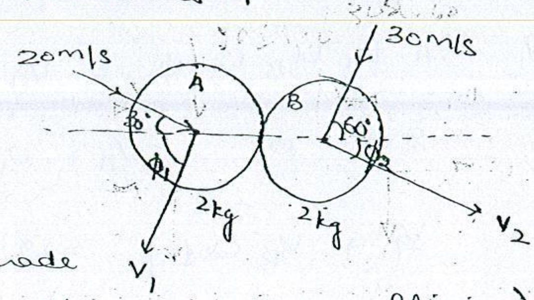
This gives $V_B = \frac{1.8}{\cos 73.3} = \frac{1.8}{.287} = \underline{6.26 \text{ m/s}}$ Ans

Putting value of $V_B \cos\phi$ in (ii) we get

$$V_A = \underline{0.2 \text{ m/s}} \rightarrow \text{ Ans}$$

... of mass 2 kg, moving with a velocity of 20 m/s strikes on a ball of mass 2 kg moving with a velocity of 30 m/s. At the instant of impact, the velocities of balls are inclined at an angle of 30° and 60° to the line joining their centres, as shown in fig. If e = 0.9 find magnitude & direction of velocities of both balls after impact

Solⁿ: Let suffix 1 & 2 refer to balls A & B respectively.



then $m_1 = m_2 = 2\,kg$

$u_1 = 20\,m/s$ & $u_2 = 30\,m/s$

$\theta_1 = 30°$, $\theta_2 = 60°$ (angles made by balls A & B resp. with line of impact before collision)

$e = 0.9$

Let ϕ_1 & ϕ_2 be the angles made by balls resp. with LOI after impact

The vertical components w.r.t. LOI of each ball before & after impact are same.

∴ For Ball A

$$V_1 \sin\phi_1 = 20 \sin 30° = 10 \quad -(i)$$

and For Ball B

$$V_2 \sin\phi_2 = 30 \sin 60° = 25.98 \quad -(ii)$$

Applying the Law of conservation of momentum along Line of impact for the system of bodies:

$$m_1 u_1 \cos\theta_1 + m_2 u_2 \cos\theta_2 = m_1 V_1 \cos\phi_1 + m_2 V_2 \cos\phi_2$$

$$\Rightarrow 2 \times 20 \cos 30° + 2 \times (-30) \cos 60° = 2 \times V_1 \cos\phi_1 + 2 \times V_2 \cos\phi_2$$

$$\Rightarrow V_1 \cos\phi_1 + V_2 \cos\phi_2 = 2.32 \quad -(iii)$$

coefficient of restitution for oblique impact is given by

$$e = \frac{V_2 \cos\phi_2 - V_1 \cos\phi_1}{u_1 \cos\theta_1 - u_2 \cos\theta_2} \Rightarrow 0.9 = \frac{V_2 \cos\phi_2 - V_1 \cos\phi_1}{20 \cos 30° + 30 \cos 60°}$$

$$\Rightarrow V_2 \cos\phi_2 - V_1 \cos\phi_1 = 29.08 \quad -(iv)$$

Solving eq. (iii) & (iv) we have

$$V_2 \cos\phi_2 = 15.7$$

This gives $V_1 \cos\phi_1 = -13.38$ (substituting $V_2 \cos\phi_2$ in iii) —(V) (-ve sign indicates the opposite direction)

Dividing eq. (i) by (v)

$$\tan\phi_1 = 0.7474$$

$$\Rightarrow \phi_1 = 36.77° \quad \text{Ans.}$$

This gives $V_1 = \frac{13.38}{\cos 36.77} = 16.7\,m/s$ & $\phi_2 = \tan^{-1}\left(\frac{V_2 \sin\phi_2}{V_2 \cos\phi_2}\right) = 58.85°$ Ans

$V_2 = \frac{15.7}{\cos 58.85} = 30.35\,m/s$ Ans.

9) A sphere of mass 4 kg moving with a velocity of 5 m/s overtakes a sphere of mass 3 kg moving with 4 m/s. If a direct impact takes place find their velocity after the impact. Assume $e = 0.5$.

Sol^n: Let $m_1 = 4\ kg$, $u_1 = 5\ m/s$

and $m_2 = 3\ kg$, $u_2 = 4\ m/s$

v_1 & v_2 = ?

$$e = 0.5 = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow v_2 - v_1 = 0.5(5-4)$$

$\Rightarrow$ $v_2 - v_1 = 0.5$ — (i)

Applying conservat^n of momentum

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$\Rightarrow 4 \times 5 + 3 \times 4 = 4v_1 + 3v_2$$

$$\Rightarrow 3v_2 + 4v_1 = 32 \quad \text{— (ii)}$$

Solving eq. (i) & (ii) we get

$v_1 = 4.36\ m/s$ and $v_2 = 4.86\ m/s$ Ans.

10.) A ball of mass 1 kg moving with a velocity of 3 m/s strikes a ball of mass 5 kg moving with a velocity of 0.6 m/s in the same direction. Show that the direct^n of motion of Ist ball is reversed. Find the loss of K.E. Assume $e = 0.75$.

Sol^n:→ Let $m_1 = 1\ kg$, $u_1 = 3\ m/s$

$m_2 = 5\ kg$, $u_2 = 0.6\ m/s$

$$e = 0.75 = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow v_2 - v_1 = 1.8 \quad \text{— (i)}$$

Applying conservation of momentum we have

$$1 \times 3 + 5 \times 0.6 = v_1 + 5v_2$$

$$\Rightarrow 5v_2 + v_1 = 6 \quad \text{— (ii)}$$

Solving eq. (i) & (ii) we have

$v_1 = -0.5$ m/s and $v_2 = 1.3$ m/s

The -ve sign indicates that the final velocity of mass 1 kg is in opposite directⁿ of its initial velocity i.e. the directⁿ of motion of Ist ball is reversed.

Now

Initial K.E $= \frac{1}{2}(m_1u_1^2 + m_2u_2^2) = 5.4$ Nm

Final K.E. $= \frac{1}{2}(m_1v_1^2 + m_2v_2^2) = 4.35$ Nm.

Loss of K.E. = Initial KE − Final K.E.

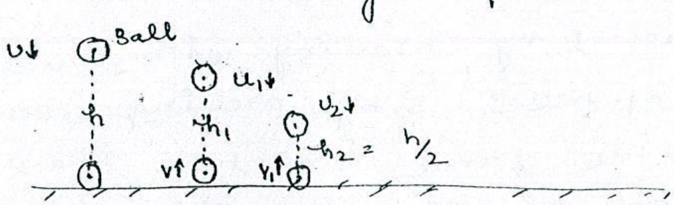
$= 5.4 - 4.35$

$= 1.05$ Nm <u>Ans</u>.

11. A ball drops from the ceiling of a room. After rebound twice from the floor it reaches a height equal to half that of ceiling. Show that coefficient of restitutⁿ is $\left(\frac{1}{2}\right)^{1/4}$.

<u>Solⁿ</u>:

Let the height of the ceiling be h.



Let the ball is dropped with initial velocity u and it rebounds with a velocity v.

The velocity of floor is zero before & after every impact.

i.e. $u_f = v_f = 0$

$$\therefore \quad e = \frac{v_f - v}{u - u_f} = -\frac{v}{u}$$

Since u & v are in opposite direction

$$e = \frac{v}{u} = \frac{\sqrt{2gh_1}}{\sqrt{2gh}} = \left(\frac{h_1}{h}\right)^{1/2}$$

$$\Rightarrow \quad h_1 = e^2 h$$

similarly

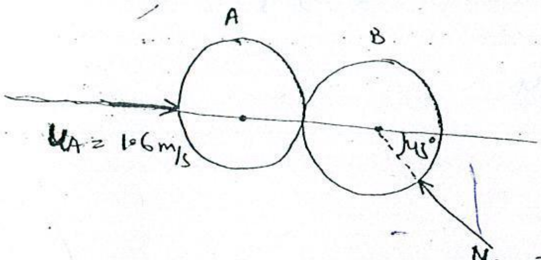
$$h_2 = e^2 h_1 = e^4 h$$

$$\Rightarrow \quad e^4 = \frac{h_2}{h} = \frac{1}{2} \qquad \left(\because h_2 = \frac{h}{2}\right) \text{ (given)}$$

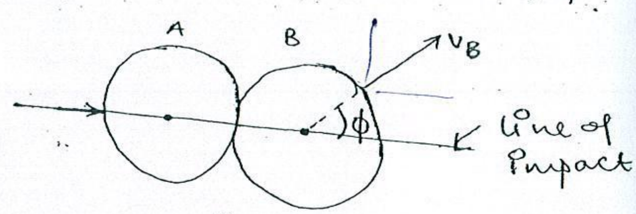
$$\Rightarrow \quad e = \left(\frac{1}{2}\right)^{1/4}$$

Hence coefficient of restitution $e = \left(1/2\right)^{1/4}$ <u>Proved</u>

12) Two identical balls each of mass M collide with velocities as shown in fig. Find the final velocities of balls after impact and loss in K.E. Assume $e = 0.9$.



Before impact



After impact

Solⁿ :→

Considering the motion of balls ~~along~~ before perpendicular to line of impact gives

$$u_B \sin 45° = V_B \sin\phi$$

$$\Rightarrow \quad V_B \sin\phi = 3.5 \times \frac{1}{\sqrt{2}} = 2.47 \quad \text{—(i)}$$

Considering motion along line of impact

$$m_A u_A + m_B(-u_B)\cos 45° = m_A V_A + m_B V_B \cos\phi$$

$$\Rightarrow \quad u_A - u_B \cos 45° = V_A + V_B \cos\phi \qquad [\because m_A = m_B = M]$$

$$\Rightarrow \quad V_A + V_B \cos\phi = 1.6 - 2.47 = -0.87 \text{ m/s} \quad \text{—(ii)}$$

Coefficient of restitution is given by

$$e = \frac{V_B \cos\phi - V_A}{u_A + u_B \cos 45°} = 0.9 \Rightarrow V_B\cos\phi - V_A = 3.66 \quad \text{—(iii)}$$

$$V_B \cos\phi - V_A = .9 \times (1.6 + 2.47) = 3.66$$

Solving eq. (ii) & (iii) gives

$$V_B \cos\phi = 1.396 \quad \text{— (iv)}$$

and $V_A = -2.27$ m/s. Ans.

Dividing (i) by (iv) we have

$$\tan\phi = \frac{2.47}{1.396} \Rightarrow \phi = 60.5^\circ$$

This gives $V_B = 2.84$ m/s

Initial value of K.E. $= \frac{1}{2}\left(m_A u_A^2 + m_B u_B^2\right)$

$$= \frac{1}{2}\left[M \times 256 + M \times (3.5)^2\right]$$

$$= 7.405\, M$$